

Lesson 2: Parameters and Return Investigate

45 minutes

Overview

In this lesson, students work with partners to investigate two different apps that use parameters and return values.

Standards

Full Course Alignment

CSP Conceptual Framework

- **AAP-2** - The way statements are sequenced and combined in a program determines the computed result. Programs incorporate iteration and selection constructs to represent repetition and make decisions to handle varied input values.
- **AAP-3** - Programmers break down problems into smaller and more manageable pieces. By creating procedures and leveraging parameters, programmers generalize processes that can be reused. Procedures allow programmers to draw upon existing code that has already been tested, allowing programmers to write programs more quickly and with more confidence.

CSTA K-12 Computer Science Standards (2017)

- **AP** - Algorithms & Programming

Agenda

Warm Up (5 minutes)

Who is Clean Code For?

Activity (35 minutes)

Wrap Up (5 minutes)

Assessment: Check For Understanding

Objectives

Students will be able to:

- Explain the benefits of using a function with a parameter or return value in the context of a specific program
- Identify situations in which a function with a parameter or return value would be necessary
- Modify programs that use functions with parameters and return
- Use the modulus operator in a program

Preparation

- Review the two apps that students will investigate in the lesson and make sure you understand the goals of the discussion prompts
- Check the "[Teacher's Lounge](#)" forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our [Virtual Lesson Modifications](#)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- [CSP Unit 7 - Lesson 2 - Parameters and Return Investigate](#) - Slides

▼ Make a Copy

Vocabulary

- **Procedural Abstraction** - a process and allows a procedure to be used only knowing what it does, not how it does it. Procedural abstraction allows a solution to a large problem to be based on the solution of smaller subproblems. This is accomplished by creating procedures to solve each of the subproblems.

Teaching Guide

Warm Up (5 minutes)

Who is Clean Code For?

 **Discuss:** *Are clean and organized programs more useful for computers or people? Why? Try to give examples from programs you've written or seen in this class.*

Have students silently journal their response, then discuss with a partner, and finally share with the class.

Discussion Goal: Parameters and return values allow students to write programs that are more organized and cleaner. Naming functions helps students write programs that read more like descriptions of what they do, and they also help students reuse code.

This quick discussion sets up many of these realizations. Students will not leave this lesson able to "do more" than previously. Instead they'll be able to write programs that are cleaner, better organized, and better able to be used elsewhere. Long term, writing programs like this will actually help students write more complex code, but because they as programmers will better be able to understand their programs and reuse the code they write. This quick discussion does not need to cover all of those points and should just be a reminder of things they've already seen. Clean code is for humans, but in the long term it helps them write better programs.

Remarks

Clean programs may sometimes run "better" but usually they're just helpful for us as we read them and write them. In other words, clean code is for people, but in the long run writing good code can help us write larger and more complicated programs because we can just read them and understand what they're doing.

Today we're going to learn more about parameters and return values. Something important to notice is that they might not actually change the way that your programs run, but they definitely will change the way we read and write programs. In general they'll make programs cleaner and easier to understand, but let's go look at some examples.

Activity (35 minutes)

 **Do This:** Take a few minutes to review the takeaways from the previous lesson.

Group: Place students with partners

 **Do This:** Students navigate to Level 1 and read and respond to the questions there. If students have extra time, they can try the modify activity.

 Teaching Tip

Make sure you leave time to bring the class back together for discussion and code walkthroughs. There's a lot to dig into with these Investigations.

- How does `calculate()` work?
- What are the arguments passed through the parameter in `calculate()` when it is called?
- What types of data does the parameter require in the `upgradeClickerCheck()` function? Where can you find that information?
- What is returned? What type of data?

 **Discuss:** As a class, walk through the questions and the code. Make sure students understand the parts of the function.

 **Do This:** Students navigate to Level 2, read lines 1-14 and discuss with their partner what's happening in these lines.

Discuss: Read through the lines together as a class, and clear up any misconceptions.

 **Do This:** Students carefully read the function on lines 15-34 before explaining to their partners how it works. If students have extra time, they can try the modify activity.

 **Discuss:** Talk through the app together as a class. Make sure students understand all the parts of the function and understand where to look in the comments for information.



1-2

App Investigation

1

2

Wrap Up (5 minutes)

 **Review:** Discuss the takeaways on the Wrap Up slide.

Discussion Goal: Students are introduced to procedural abstraction on this slide, a concept they will return to throughout the unit. Spend a few minutes discussing what "generalize" means - focusing on how a single function can be used to accomplish many different tasks by using parameters.

Assessment: Check For Understanding

Check For Understanding Question(s) and solutions can be found in each lesson on Code Studio. These questions can be used for an exit ticket.

Question: What are the benefits of writing functions that use parameters and return? Try to list at least two.



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